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Study of Flow Behaviour in Vertical Centrifugal Casting

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Abstract

In centrifugal casting processes, flow of fluid layers determines the quality of the cast. However flow visualization is not possible, as mould and molten metal are opaque in nature. Therefore many research workers have resorted to understand the flow visualization process by carrying out simulation studies. Literature survey indicates that earlier investigators have studied the melt flow in horizontal centrifugal casting process by conducting cold modeling experiment using horizontal transparent mould and fluids of different viscosity. However it appears that there is no published information available in the area of influence of various parameters in the case of vertical centrifugal casting process. Therefore in this research work the cold modeling experimentation on fluids of different viscosity like water, Glycerin and SAE 90 rotated in a transparent mould at a constant room temperature had been conducted with an objective of understanding the influence of rotational speed and aspect ratio on flow patterns of different fluids subjected to vertical axis rotation. Later the flow behavior had been studied by numerical simulation through Ansys fluent software. Further for the purpose of comparison, casts of paraffin wax at different rotational speeds had been obtained and the trend of variation of the lift with the variation of rotational speed had been compared with the trend predicted using cold modeling studies and using Ansys software. From cold modeling studies it can be concluded that of the dimensional accuracy of fluid cylinder in terms of lift had been increased with increase in viscosity. The possibility of formation of complete fluid cylinder increases with increase in rotational speed and mould diameter. Howvere, at aspect ratio of 1.0 the formation of complete fluid cylinder had not been obtained.The trend of result observed in CFD study was found to be similar to the trend of results observed in cold modeling study. The study of actual paraffin cast confirmed the result obtained in cold modeling and CFD.

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Keywords: Vertical Centrifugal Casting:Cold Modeling;Rotational Speed;ANSYS Fluent

1.Introduction

The centrifugal casting process consists of pouring the molten metal at a suitable temperature into a rapidly rotating mould. Depending upon the axis of rotation of the mould either horizontal or vertical,the process has been termed as horizontal centrifugal or vertical centrifugal casting process. In centrifugal casting processes, flow of fluid layers determines the quality of the cast. However, flow visualization is not possible as mould and molten

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metal are opaque in nature.Therefore many research workers have resorted to understand the flow visualization process by carrying out simulation study[1] to[3].

Yogesh Jaluria[5]. has made a detail study on fluid flow phenomenon in material processing and has characterized the parameters affecting the centrifugal casting process.Mukunda et.al.[6] have carried out experimental studies of flow patterns of different fluids in partially filled rotating cylinder. They have reported that rotational speed plays a major role in obtaining a sound cast. With increase in rotational speed, different types of flow like Marangoni flow, Couette flow, and Taylor flow, will be generated, and there exists a critical value at which a complete cylinder is formed . Viscosity is another important factor that decided the flow pattern of the melt. It was observed that in case of horizontal centrifugal casting the formation of a liquid cylinder occurs at higher rotational speed for fluids having lower viscosity, whereas ,in case of fluids having higher viscosity, the liquid cylinder is formed at a lower rotational speed .Similarly, if aspect ratio is increased, Ekman flow is not significant, but the formation of a complete hollow fluid cylinder in large diameter moulds takes place at reduced rotational speed. Further, in the case of thicker fluids, cylinder formation is easy and is obtained at a lower rotational speed.Keerthi etal [6]has analyzed the fluid flow in horizontal centrifugal casting process .Further they[7] have also numerically simulated the cold modeling experiments and have studied the type of cylinder formed .Zagorski et.al[8] have studied mould pouring phenomenon during horizontal centrifugal casting process by simulation process.

In horizontal casting process .the molten metal when poured to rotating mould the pick up time is more, as gravitational force pulls the melt in down ward direction . Vertical centrifugal casting is a type of casting process in which molten metal is allowed to rotate about a vertical axis to form cylindrical shapes where the aspect ratio is as minimum as possible. This technique is being applied to manufacture rings, bearings and also to produce non-cylindrical shaped parts such as valves, propellers, etc. Unlike horizontal centrifugal casting, the resultant force on the liquid in vertical centrifugal casting will remain constant. Hence molten metal is picked up along with the mould as soon as poured into the rotating mould[9].Smrutirekha etal. [9,10] in their investigation have studied the flow behaviour by obtaining cast of paraffin wax in horizontal casting process. Above literature survey indicates that earlier investigators have studied the melt flow in horizontal centrifugal casting process by conducting cold modeling experiment using horizontal transparent mould and fluids of different viscosity. However it appears that there is no published information available in the area of influence of various parameters in the case of vertical centrifugal casting process. Therefore in this research work it has been proposed to carry out a systematic study to visualize the flow in vertical centrifugal casting process.It has been proposed to conduct the cold modeling experimentation on permanent fluid of different viscosity like water, Glycerin and SAE 90 rotated in a transparent mould at a constant room temperature with an objective of understanding the influence of rotational speed and aspect ratio on flow patterns of different fluids subjected to vertical axis rotation.

Later it has been proposed to compare the flow behavior obtained by numerical simulation through Ansys fluent software. Further it has been proposed to produce casts of paraffin wax at different rotational speeds and observe the trend of variation of the lift with the variation of rotational speed and compare with the trend predicted using cold modeling studies and using Ansys software. While carrying out the experiments, it has been assumed that physical properties of the melt will be retained in liquid phase and the thermal aspects are negligible and the details of experimentation results of the systematic study carried out in above lines has been reported in this paper

2. Experimental Details

2.1. Cold Modeling study

The cold modelling and vertical centrifugal casting set ups shown in figure 1 and 2 are indigenously fabricated. These equipments consist of 3-phase AC/ DC motor, autotransformer, mould and proximity sensor. The variation in rotation speed of mould possible is from 20 to 2000rpm. The proximity sensor is used to note the number of rotation of mould. The fluid properties and the dimension of mould is presented in Table 1 and Table 2.

Table 1: Fluid properties

Fluid	Density in g/cc	Kinematic viscosity in mm^2/sec
Water	1.0	0.806
SAE 90	0.892	13.13
Glycerin	1.26	648

Table 2:Dimensions of Mould with different aspect ratios considered for the study

Mould Diameter in mm(D)	Thickness of Fluid cylinder in mm	Mould length in mm(L)	Aspect Ratio (L/D)
60	4	30	0.5
		60	1.0
90	4	45	0.5
		90	1.0
120	4	60	0.5
		120	1.0

Figure 1. Cold modeling Setup

The volume of fluid required to form a fluid cylinder of 4mm thickness was calculated for different mould dimensions. The required volume of fluid was poured into the rotating mould at a definite speed. The flow patterns were observed at three different rotational speed i.e 300rpm,400rpm and 500rpm.The time taken for the fluid to reach a stable state (i.e. formation of a fluid cylinder or fluid reaching the extreme end of the mould) was recorded, which included pouring time as well as lift time. The effect of factors like aspect ratio, viscosity of the fluid on fluid flow behaviour at constant rotational speed was observed.

2.2 CFD Analysis

In the present work, to perform the Computational Fluid Dynamics (CFD) analysis, ANSYS Fluent software has been chosen, which is being used for validation in case of multiphase and rotational flows. The intention to choose CFD analysis was to compare the experiments conducted and to predict the cylinder thickness, cylinder height and optimize the mould rotation speed. The fluid was considered to be under the action of centrifugal force along the radial direction as well as gravitational force along the downward direction

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\partial v_x) + \frac{\partial}{\partial r}(\partial v_r) + \frac{\partial v_r}{r} = S_m \qquad (1)$$

The equation expressed above is meant for solving the continuity equation for 2D axisymmetric geometries. Where x is the axial coordinate, r is the radial coordinate, v_x is the axial velocity and v_r is the radial velocity. However, the continuity equation for multiphase flows has been modified as given below. The VOF model (described in Volume of Fluid (VOF) Model Theory) a form of surface tracking technique has been applied to a fixed Eulerian mesh. In the VOF model, a single set of momentum equations is shared by the fluids, and the volume fraction of each of the fluids in each computational cell is tracked throughout the domain. Hence the flow patterns that were observed for the 2D model confirm the cylinder dimension. The VOF model considered for the present work is shown in figure 2.

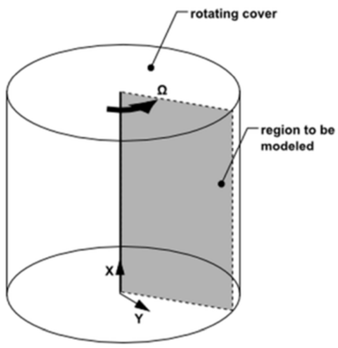


Figure 2 VOF Model

In the present investigation with respect to the real experimentation, vertical rotating cylinder is taken in the pre processing phase of CFD analysis; the experimental model was decomposed into a 2-dimensional symmetric model to avoid the computational expenses and time. For the solution phase, a CAD model was discretized into quadrilateral shape elements and the solution was based on 2D symmetric model satisfying flow equations along with multiphase equations, as air and liquid has been considered as two fluids. Similar to experiments on cold modelling, rotational speeds of the mould considered for CFD was 300rpm,400 rpm and 500rpm.The time taken for formation of a fluid cylinder as recorded in the cold modeling experiments has been considered in the analysis as well to validate flow physics of different fluids. CFD simulation with experimental results has been compared for the height of the fluid cylinder during the rotation of the mould and the shape of the fluid cylinder at the end of the flow time.

2.3 Study of Actual Cast

In order to study flow behaviour of melt, paraffin wax has been subjected to vertical centrifugal casting process at different rotational speed. The centrifugal casting set up shown in figure 3, consists of mild steel mould fixed to a flange which is connected to the shaft of a 1HP DC shunt motor whose speed can vary from 20 rpm to 2000 rpm.

Figure 3 Centrifugal Casting Set up

In this process, the molten metal was poured into a vertically rotating mould , which is subjected to a centrifugal force that act in the outward direction that result in the formation of hollow cylinder with required thickness.

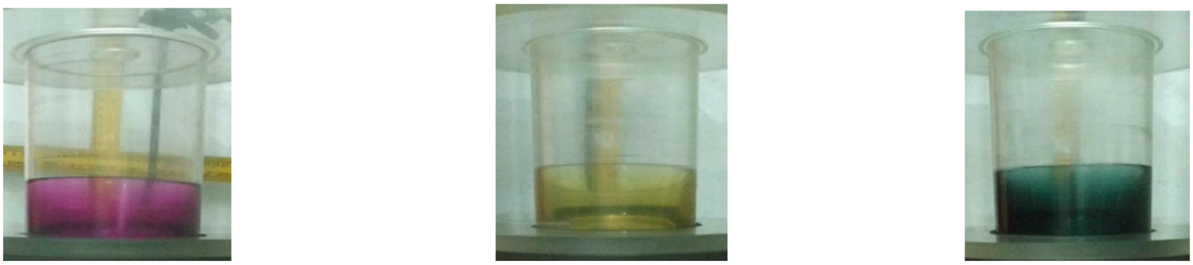
3. Results and Discussions

3.1 Results Pertaining to Cold Modeling Study

The formation of fluid cylinder observed for three different fluids for different mould diameter at 300rpm are represented in figure 4-6 .When fluid was introduced into rotating mould, it gets lifted in upward direction instantaneously forming a conical cylinder with some tapering angle or sometimes forming a complete fluid cylinder. From figure 4, it can be observed that at rotational speed of 300rpm in mould of diameter 120mm and aspect ratio 0.5 the taper angle of the fluid cylinder thus formed is maximum, minimum and intermediate in case of water Glycerin and SAE 9 respectively. The same trend can be observed from figure 5 and figure 6 in case of mould of diameter 90mm and 60mm.However at rotational speed of 300rpm, required diameter and length of fluid cylinder was not achieved in any of the fluid and with any mould diameter considered for the present work. With increase in rotational speed the dimensional accuracy of the fluid cylinder was found to be improved. By comparing figure 4 - 6 it can be observed that fluid cylinder formed in case of mould of diameter 120mm is more dimensionally accurate as compared to mould of diameter 90mm and 60mm because of larger G force. A comparison of G force is made in figure 7. The lift of the fluid cylinder with respect to rotational speed is represented in form of graph in figure 8.

a. Water b.SAE 90 c. Glycerin

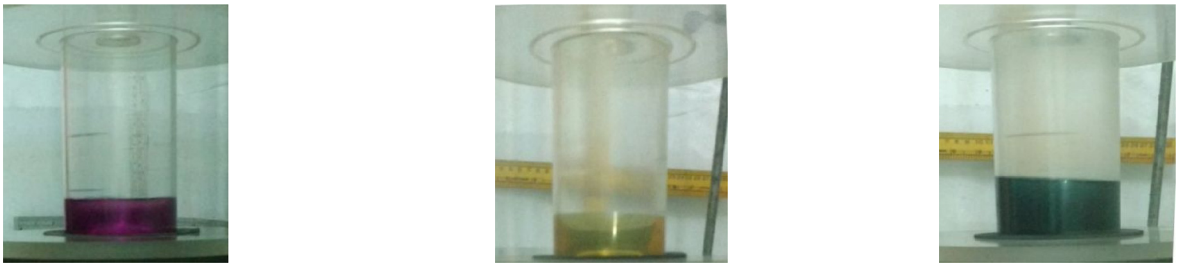
Figure 4 Formation of fluid cylinder at aspect ratio 0.5 in rotating mould of diameter 120mm at 300rpm



a. Water b.SAE 90 c. Glycerin

Figure 5 Formation of fluid cylinder at aspect ratio 0.5 in rotating mould of diameter 90mm at 300rpm

Thus within the scope of the investigation, the lift of the fluid cylinder formation is low in case of water or in other words, the lift of the fluid is low in case of liquid having lesser viscosity. The lift of fluid cylinder was found to increase with increase in viscosity. Hence with the increase in Kinematic viscosity of fluid, increase in lift was found to be exponential in nature. This trend of observation has been found true in all cases of aspect ratio, mould diameter, rotational speed considered. Unlike, an observation of Taylor-Couette flow in horizontal centrifugal casting, while in vertically rotating mould raining effect was recorded in case of water at lower rotational speed. Also it is noted that the fluid cylinder obtained in higher mould diameter has high dimensional accuracy.



a. Water b.SAE 90 c. Glycerin

Figure 6. Formation of fluid cylinder at aspect ratio 0.5 in rotating mould of diameter 60mm at 300rpm

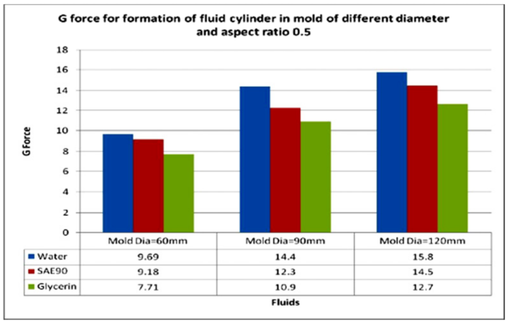


Figure 7. Comparison of G Force for formation fluid cylinder in different mould diameter for thickness 4mm and aspect ratio 0.5

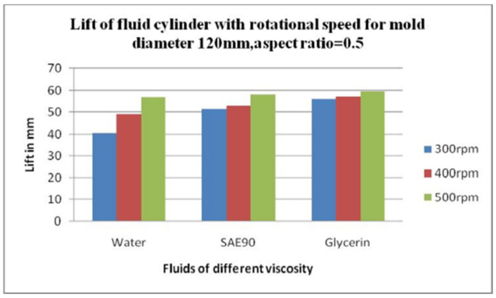
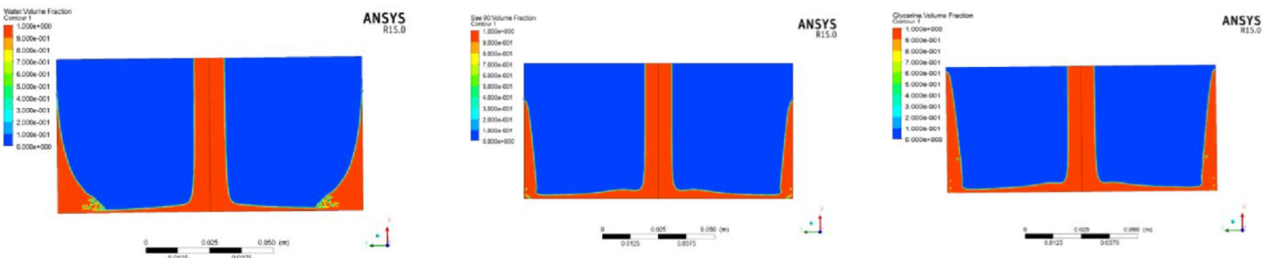


Figure 8. Lift of fluid cylinder with rotational speed for mould of diameter 120mm and aspect ratio 0.5

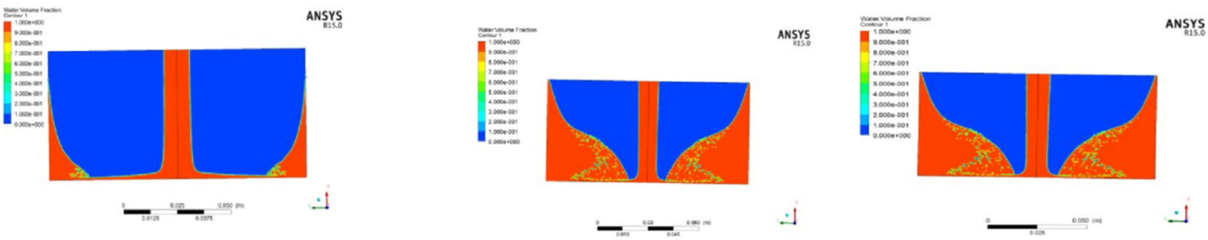
3.2 Results Pertaining to CFD Study

The formation of fluid cylinder for different fluids at typical aspect ratio 0.5 for the mould of 120mm at 300rpm is shown in figure 9.it can be observed that the lift of the fluid cylinder is maximum, minimum and intermediate Glycerin, water and SAE90 respectively which also confirm the result of cold modelling experiments. This indicate that lift of the fluid is directly proportional to the viscosity of the fluid.



a. Water b.SAE90 c.Glycerin

Figure 9. Formation of fluid cylinder for different fluids at a typical aspect ratio 0.5, mould diameter 120mm and rotational speed of 300 rpm



a.300rpm b. 400rpm c.500rpm

Figure 10. Formation of fluid cylinder for different fluids in case of water at a typical aspect ratio 0.5, mould diameter 120mm at different rotational speeds

From figure 10 it can be observed that lift of the fluid cylinder increases with increase in rotational speed. Also the raining effect of water is clearly visualized. From Table 3, a comparison of lift of fluid cylinder obtained in cold modeling and CFD analysis has been made. The values obtained in CFD confirms the trend of results obtained in cold modeling. The well established VOF model theory was found relevant to the results of the research work obtained in the cold modeling.

Table 3 Lift of fluid cylinder

Mould Diameter in mm	Rotational Speed in rpm	Types of Fluids	Lift of fluid cylinder in Cold modeling in mm	Lift of fluid cylinder in CFD
60	300	Water	25	27.54
		SAE90	27	28.3
		Glycerin	28	28.6
90	300	Water	40	40.1
		SAE90	45	45
		Glycerin	47	45
120	30	Water	42	47.3
		SAE90	44	50.2
		Glycerin	45	56.11

3.3. Results pertaining to actual cast

In order to compare the results of cold modeling and CFD analysis, cast of paraffin wax were obtained at 300rpm,400rpm and 500rpm as shown in figure 11 .It can be observed that the dimensional accuracy of the cast improves with increase in rotational speed. The dimensional accuracy in terms of height of the typical cast of paraffin wax obtained at different rotational speed has been represented in Table 4

a.300rpm b.400rpm c.500rpm

Figure 11 Cast of Paraffin wax obtained at different rotational speed

Also with increase in rotational speed the tapering angle of the cylinder at the inner surface reduces as a result of which cylinder with desired thickess, diameter and height was found to be achieved at rotational speed 0f 500rpm.Also the surface finish of the cast obtained was found to be best in case 500rpm compared to 400rpm and 300rpm respectively, which can be clearly visualized .

Table 4 -Dimensional accuracy of typical cast of paraffin wax at different rotational speeds

Rotational speed	Desired dimension in mm	Actual dimension in mm	Deviation %	Remarks
300rpm	45	40	11.1	Poor dimensional accuracy achieved
400rpm	45	43	4.44	Moderate dimensional accuracy achieved
500rpm	45	45	NIL	Good dimensional accuracy achieved

4. Conclusions

Based on the results of cold modeling studies, it has been found that the lift of the fluid cylinder is increased with increase in viscosity of fluid. The dimensional accuracy of the fluid cylinder is found to be better in case of fluid of higher viscosity. The possibility of formation of complete fluid cylinder increases with increase in rotational speed. With increase in mould diameter, G force increases which result in formation of fluid cylinder of better dimensional accuracy. An aspect ratio of 1.0 does not provided the formation of complete fluid within the scope of investigation. Further the trend of results observed in CFD study was similar to those observed in cold modelling study. The study of actual paraffin cast confirms the result obtained in cold modelling and CFD.

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